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Long-Term Health Outcomes of Traumatic Brain Injury in Veterans

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Traumatic brain injury (TBI) has become one of the most prevalent battle wounds among veterans of the Iraq and Afghanistan wars. According to the Defense and Veterans Brain Injury Center,[1] nearly 414 000 service members worldwide sustained TBI between 2000 and late 2019. More than 185 000 veterans receiving Veterans Affairs (VA) health care have received at least 1 diagnosis of TBI, mostly mild cases. The impact of TBI and related conditions substantially burdens service members, families, clinicians, and health systems.[1]

Numerous psychiatric and medical consequences arise after TBI, including posttraumatic stress disorder, depression, suicidal thoughts, cognitive deficits, chronic pain, and unemployment. TBI severity appears to be linked to certain outcomes. Given this backdrop, the analysis by Stewart et al[2] provides a timely probe of associations between TBI severity and subsequent risk of brain cancer in this vulnerable patient cohort. This retrospective cohort study analyzed records of more than 1.9 million post-9/11 veterans. Notable findings indicate that moderate, severe, and penetrating TBI were associated with elevated brain cancer risk, but mild TBI was not. This finding suggests that severe TBI may predispose veterans to brain cancer, whereas mild TBI likely is not associated with heightened risk. Crude incidence rates per 100 000 person-years were 3.06 for no TBI, 2.85 for mild TBI, 4.88 for moderate/severe TBI, and 10.34 for penetrating TBI ($P < .001$), displaying a stepwise increase in brain cancer incidence corresponding to worse TBI severity.[2]

Additional studies highlight gaps in cancer surveillance data for veterans. An analysis by Woo et al[3] found 25% of Ohio veterans with a primary brain tumor were omitted from the state cancer registry. Direct data matching revealed more glioblastomas in veterans vs nonveterans, independently of sex, exemplifying how veteran-specific cancer patterns may diverge from population-level figures. Although definitive causal links between military environmental and/or chemical exposures and subsequent illness remain undetermined, the recent Sergeant First Class Heath Robinson Honoring our Promise to Address Comprehensive Toxics (PACT) Act[4] expands care access and benefits for veterans reporting toxic exposures. Because veterans comprise 7% of the US adult populace,[5] with half obtaining care via the VA, elucidating contributors to brain cancer in this cohort is critical. Brain tumors confer substantial morbidity and mortality, underscoring the value of the authors' research.[2]

The PACT Act[4] makes several key provisions supporting veterans with brain tumors and other potentially exposure-related illnesses. According to recent VA and National Institutes of Health data,[6] glioblastoma, the most aggressive malignant brain tumor, is the third leading cause of cancer-related death among active-duty personnel. Post-9/11 veterans deployed to Iraq, Afghanistan, and elsewhere face a 26% higher glioblastoma rate vs the general public, with average age of onset decades earlier than in broader populations.[6] Beyond expanding health care access, the PACT Act will heighten research into potential connections between toxic exposures and early-onset cases in younger veterans. Additional funding and coordination could help collect vital data to clarify brain tumor patterns, causes, and risk factors in this group. An improved scientific understanding of links between exposures and illness may ultimately help reduce brain tumor burden and mortality among veterans.

As we continue disentangling connections between combat injury and long-term health, a promising research avenue examines relationships between specific neurotrauma biomechanics and subsequent neurodegeneration. Assessment of TBI biomechanics using computational modeling tools such as finite element analysis facilitates replicating and evaluating resulting brain damage that

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otherwise would be difficult to access. Such simulations aid injury diagnosis and early identification of long-term damage, thereby promoting preventative measures and timely treatment.

Sophisticated finite element models incorporating advanced constitutive equations and damage models can simulate complex TBI biomechanics. By accounting for features like time-rate effects, nonlinear tissue behaviors, and irreversible damage from impact waves or other loads, multilayered finite element models can replicate persisting injury.[7-9] Characteristics facilitating diffuse axonal injury and cavitation damage simulation thereby enable elucidating subclinical and clinical consequences. Further incorporation of exact finite viscoelasticity theories producing substantial viscous deformations and perturbations from equilibrium may help answer pressing questions by analyzing relationships between biomechanical parameters and injury.

Overall, this study[2] provides meaningful data clarifying associations between combat-related TBI severity and subsequent brain cancer risk among post-9/11 veterans. Elucidating potential connections between battlefield trauma and longer-term health outcomes is imperative to inform prevention and care approaches for those who have served. In particular, because the PACT Act expands access and benefits for veterans affected by illness that may relate to hazardous exposures, understanding contributors is critical.

In addition, an improved scientific understanding of how traumatic injury forces propagate through brain tissue to trigger neurodegeneration and other pathological conditions may shed further light on the observed links between TBI and elevated brain cancer rates. Research illuminating biomechanical underpinnings of posttraumatic sequelae will enable optimization of screening, diagnosis, and treatment. For the growing number of younger veterans facing cancer, clarifying potential intersections between trauma mechanisms, toxic exposures, and cancers through PACT Act–enabled research is instrumental in protecting long-term well-being after battle.

ARTICLE INFORMATION

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